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Devices and techniques for performing an osteotomy procedure on a first metatarsal to correct a bone misalignment

Abstract

Devices and techniques for adjusting an alignment for a first metatarsal may include making a plurality of cuts along the length of the first metatarsal. In different applications of the technique, a plurality of transverse cuts may or may not be made intersecting the longitudinal cuts. In either case, the cuts can separate the first metatarsal into two individual portions and release a removable wedge of bone. The different portions of the metatarsal can be moved relative to each other to adjust an anatomical alignment of one portion relative to another portion, for example in multiple planes. After suitably adjusting the alignment of the different bone portions, the portions can be fixed together.

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Background/Summary

CROSS-REFERENCE (1) This application is a continuation of U.S. patent application Ser. No. 16/812,487, filed Mar. 9, 2020, and issued as U.S. Pat. No. 11,364,037 on Jun. 21, 2022, which is a continuation of U.S. patent application Ser. No. 15/809,298, filed Nov. 10, 2017, and issued as U.S. Pat. No. 10,582,936 on Mar. 10, 2020, which claims priority to U.S. Provisional Patent Application No. 62/421,027, filed Nov. 11, 2016. The entire contents of each of these applications are incorporated herein by reference.

TECHNICAL FIELD

(1) This disclosure relates to devices and techniques for correcting bones and, more particularly, to osteotomy techniques for correcting bone misalignment.

BACKGROUND

(2) Bones, such as the bones of a foot, may be anatomically misaligned. In certain circumstances, surgical intervention is required to correctly align the bones to reduce patient discomfort and improve patient quality of life.

SUMMARY

(3) In general, this disclosure is directed to devices and techniques for correcting an anatomical misalignment of one or more bones. In some examples, the devices and techniques are utilized to correct a misalignment of a first metatarsal relative to a medial cuneiform and/or an adjacent second metatarsal, such as bunion correction procedure. To perform a corrective procedure, a clinician may make two parallel but offset cuts along the length of the first metatarsal. The two cuts may be angled relative to each other across the thickness of the bone, causing the cuts to intersect to form a wedged-shaped section of bone. If the cuts do not extend the entire length of the bone, the clinician can further make two transverse cuts on opposite ends of the longitudinal cuts, e.g., adjacent the proximal and distal ends of the first metatarsal, to release the wedge-shaped section of bone from a remainder of the metatarsal. The transverse cuts can be made before, after, or between making each of the two longitudinal cuts.

(4) Upon resecting the wedge-shaped section of bone from the first metatarsal, the first metatarsal

may be divided into two independently movable portions. In cases where the first metatarsal is cut with longitudinal and transverse cuts, the metatarsal may be divided into a proximal portion connected to the medial cuneiform and a distal portion connected to the proximal phalanx. In cases where the first metatarsal is along the longitudinal axis but without transverse cuts, the metatarsal may be divided into a plantar portion and a dorsal portion. In either case, clinician can move the two portions of the first metatarsal relative to each other in multiple planes to help correct the anatomical misalignment of the first metatarsal (e.g., the distal portion of the first metatarsal). For example, the clinician may rotate one portion of the first metatarsal in the frontal plane, pivot the portion of the first metatarsal in the transverse plane, and/or pivot the portion of the first metatarsal in the sagittal plane to adjust the anatomical alignment of the one portion of the first metatarsal relative to the other portion of the first metatarsal.

(5) In some applications, the clinician removes the wedge portion cut from the first metatarsal to provide clearance for realignment. This can enable one portion of the metatarsal (e.g., the distal portion) to be rotated relative to the other portion of the first metatarsal (e.g., the proximal portion). In some examples, a clinician may also trim around the perimeter of the cut end of the distal portion and/or proximal portion of the first metatarsal, e.g., to avoid causing the rotated bone portion to create an interfering lip or edge where it is rotated out of plane.

(6) The clinician may or may not use the wedge-shaped bone portion resected from the first metatarsal as an autograft to fill a space created during realignment of one portion of the cut first metatarsal relative to another portion of the cut metatarsal. For example, the wedge-shaped bone portion may be resected from the medial side of the first metatarsal to create a space that allows for realignment of a first portion of the metatarsal relative to a second portion of the metatarsal. For example, a distal portion of the first metatarsal may be moved in one or more planes (e.g., the frontal plane) relative to the proximal portion, closing the space created by removal of the wedge-shaped bone portion. A corresponding gap may be created on the lateral side of the first metatarsal through realignment. The wedge-shaped bone portion can be inserted as an autograft in this gap. The clinician may or may not trim or otherwise resize the wedge-shaped bone portion to fit within the space created through realignment. Instead of reusing the bone portion harvested from the first metatarsal as an autograft, the clinician may use a different bone construct to fill the gap created through realignment, such as an allograft harvested from another person, a xenograft harvested from a different species, or synthetic bone.

(7) Although different instruments may be used to perform the bone correction procedure, in some examples, a bone cutting guide is used that has a longitudinal cutting slot. In one example, the bone cutting guide has a longitudinal cutting slot and a plurality of fixation apertures to allow the guide to be fixated at different rotational and angular positions about the first metatarsal to be cut. Additionally or alternatively, the bone cutting may have one or more transverse cutting slots that extend upwardly and/or downwardly from the longitudinal cutting slot. For example, the bone cutting guide may include a longitudinal cutting slot, a first transverse cutting slot that intersects one end of the longitudinal cutting slot, and a second transverse cutting slot that intersects the opposite end of the longitudinal cutting slot. In use, a clinician may use the longitudinal cutting slot to cut along the length of the first metatarsal and, after rotating the cutting guide, use the longitudinal cutting slot to make a second, intersecting cut along the length of the first metatarsal. The clinician may further use the two transverse cutting slots to make intersecting cuts to the two longitudinal cuts (e.g., in the dorsal to plantar direction, or vice versa). This may cleave the first metatarsal into adjacent proximal and distal portions for realignment.

(8) In one example, a method is described that includes making a first longitudinal cut through a first metatarsal between a proximal end of the first metatarsal and a distal end of the first metatarsal. The method also includes making a second longitudinal cut through the first metatarsal between a proximal end of the first metatarsal and a distal end of the first metatarsal, where the second longitudinal cut is radially offset from the first longitudinal cut and intersects the first

longitudinal cut. The method further involves making a transverse cut adjacent the proximal end of the first metatarsal that intersects the first longitudinal cut and the second longitudinal cut and making a transverse cut adjacent the distal end of the first metatarsal that intersects the first longitudinal cut and the second longitudinal cut. In addition, the method includes removing a bone wedge from the first metatarsal and moving a distal portion of the first metatarsal relative to a proximal portion of the first metatarsal in at least two planes, thereby adjusting an anatomical alignment of the distal portion of the first metatarsal relative to the proximal portion of the first metatarsal.

(9) In another example, a bone cutting guide is described that includes a body configured to be positioned against a bone to be cut. The body includes a longitudinal cutting slot having a first end and a second end, a first transverse cutting slot extending downwardly from the first end of the longitudinal cutting slot, and a second transverse cutting slot extending upwardly from the second end of the longitudinal cutting slot.

(10) In another example, a bone cutting guide is described that includes a body having a length extending from a first end to a second end and that is sized to be positioned against a first metatarsal. The example specifies that the body includes a longitudinal cutting slot extending between the first end and the second end, a first securing projection extending substantially orthogonally relative to the length in one direction and containing at least one fixation aperture, and a second securing projection extending substantially orthogonally relative to the length in an opposite direction and containing at least one fixation aperture.

(11) In another example, a method is described that includes making a first longitudinal cut through a first metatarsal extending from a proximal end of the first metatarsal to a distal end of the first metatarsal and making a second longitudinal cut extending from the proximal end of the first metatarsal to the distal end of the first metatarsal. The example specifies that the second longitudinal cut is radially offset from the first longitudinal cut and intersects the first longitudinal cut, thereby separating the first metatarsal into a first portion, a second portion, and a bone wedge. The technique involves removing the bone wedge from the first metatarsal and moving the first portion of the first metatarsal relative to the second portion of the first metatarsal, thereby adjusting an anatomical alignment of the first portion of the first metatarsal relative to the second portion of the first metatarsal.

(12) The details of one or more examples are set forth in the accompanying drawings and the description below. Other features, objects, and advantages will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIGS. 1A and 1B are front views of a foot showing a normal first metatarsal position and an example frontal plane rotational misalignment position, respectively.

(2) FIGS. 2A and 2B are top views of a foot showing a normal first metatarsal position and an example transverse plane misalignment position, respectively.

(3) FIGS. 3A and 3B are side views of a foot showing a normal first metatarsal position and an example sagittal plane misalignment position, respectively.

(4) FIG. 4 is a flow diagram illustrating an example osteotomy technique for correcting an anatomical misalignment.

(5) FIG. 5A is sectional image of a first metatarsal taken from the frontal plane showing example cut lines that can be made while performing the technique of FIG. 4.

(6) FIG. 5B is a perspective image of the example first metatarsal from FIG. 5A.

(7) FIG. 5C is a sectional view of an example first metatarsal showing an example use of a wedge-

shaped bone portion to close a gap created during realignment.

(8) FIG. 5D is a dorsal perspective view of the example first metatarsal with wedge-shaped bone portion from FIG. 5C.

(9) FIGS. 5E and 5F are a medial side view and a frontal plane view, respectively, of a first metatarsal illustrating an example technique in which the metatarsal is separated into two portions by making two cuts along the length of the metatarsal.

(10) FIGS. 5H and 5G are a medial side view and a frontal plane view, respectively, showing an example use of wedge-shaped bone portion in a metatarsal realignment technique.

(11) FIG. 6A is a perspective view of one example cutting guide that can be used to cut a first metatarsal to realign a distal portion of the metatarsal relative to a proximal portion of the metatarsal.

(12) FIG. 6B is a perspective view of the example cutting guide of FIG. 6A shown attached to a first metatarsal.

(13) FIG. 7A is a perspective illustration of another example configuration of a bone cutting guide that can be used to cut a first metatarsal to realign a distal portion of the metatarsal relative to a proximal portion of the metatarsal.

(14) FIG. 7B is a perspective illustration of the example cutting guide of FIG. 7A shown attached to a first metatarsal.

(15) FIG. 8 is a perspective illustration of another example bone cutting guide that can be used according to the disclosure.

(16) FIGS. 9A-9D are conceptual illustrations showing how the example bone cutting guide of FIG. 8 can be manipulated relative to a bone to be cut to execute a bone realignment technique.

DETAILED DESCRIPTION

(17) In general, the present disclosure is directed to devices and techniques for correcting a misalignment of one or more bones. The disclosed devices and techniques can be implemented in an osteotomy procedure in which a bone is surgically cut and/or a piece of bone is surgically removed. In some examples, the technique is performed on one or more bones in the foot or hand, where bones are relatively small compared to bones in other parts of the human anatomy. For example, the foregoing description generally refers to example techniques performed on the foot and, more particularly a metatarsal of the foot. However, the disclosed techniques may be performed on other bones, such as the tibia, fibula, ulna, humerus, femur, or yet other bone, and the disclosure is not limited in this respect unless otherwise specifically indicated. In some applications, however, the disclosed techniques are used to correct a misalignment between a metatarsal (e.g., a first metatarsal) and a second metatarsal and/or a cuneiform (e.g., a medial, or first, cuneiform), such as in a bunion correction surgery.

(18) FIGS. 1-3 are different views of a foot 200 showing example anatomical misalignments that may occur and be corrected according to the present disclosure. Such misalignment may be caused by a hallux valgus (bunion), natural growth deformity, or other condition causing anatomical misalignment. FIGS. 1A and 1B are front views of foot 200 showing a normal first metatarsal position and an example frontal plane rotational misalignment position, respectively. FIGS. 2A and 2B are top views of foot 200 showing a normal first metatarsal position and an example transverse plane misalignment position, respectively. FIGS. 3A and 3B are side views of foot 200 showing a normal first metatarsal position and an example sagittal plane misalignment position, respectively. While FIGS. 1B, 2B, and 3B show each respective planar misalignment in isolation, in practice, a metatarsal may be misaligned in any two of the three planes or even all three planes. Accordingly, it should be appreciated that the depiction of a single plane misalignment in each of FIGS. 1B, 2B, and 3B is for purposes of illustration and a metatarsal may be misaligned in multiple planes that is desirably corrected.

(19) With reference to FIGS. 1A and 2A, foot 200 is composed of multiple bones including a first metatarsal 210, a second metatarsal 212, a third metatarsal 214, a fourth metatarsal 216, and a fifth

metatarsal **218**. The metatarsals are connected distally to phalanges **220** and, more particularly, each to a respective proximal phalanx. The first metatarsal **210** is connected proximally to a medial cuneiform **222**, while the second metatarsal **212** is connected proximally to an intermediate cuneiform **224** and the third metatarsal is connected proximally to lateral cuneiform **226**. The fourth and fifth metatarsals **216**, **218** are connected proximally to the cuboid bone **228**. The joint **230** between a metatarsal and respective cuneiform (e.g., first metatarsal **210** and medial cuneiform **222**) is referred to as the tarsometatarsal (“TMT”) joint. The joint **232** between a metatarsal and respective proximal phalanx is referred to as a metatarsophalangeal joint. The angle **234** between adjacent metatarsals (e.g., first metatarsal **210** and second metatarsal **212**) is referred to as the intermetatarsal angle (“IMA”).

(20) As noted, FIG. **1A** is a frontal plane view of foot **200** showing a typical position for first metatarsal **210**. The frontal plane, which is also known as the coronal plane, is generally considered any vertical plane that divides the body into anterior and posterior sections. On foot **200**, the frontal plane is a plane that extends vertically and is perpendicular to an axis extending proximally to distally along the length of the foot. FIG. **1A** shows first metatarsal **210** in a typical rotational position in the frontal plane. FIG. **1B** shows first metatarsal **210** with a frontal plane rotational deformity characterized by a rotational angle **236** relative to ground, as indicated by line **238**.

(21) FIG. **2A** is a top view of foot **200** showing a typical position of first metatarsal **210** in the transverse plane. The transverse plane, which is also known as the horizontal plane, axial plane, or transaxial plane, is considered any plane that divides the body into superior and inferior parts. On foot **200**, the transverse plane is a plane that extends horizontally and is perpendicular to an axis extending dorsally to plantarly (top to bottom) across the foot. FIG. **2A** shows first metatarsal **210** with a typical IMA **234** in the transverse plane. FIG. **2B** shows first metatarsal **210** with a transverse plane rotational deformity characterized by a greater IMA caused by the distal end of first metatarsal **210** being pivoted medially relative to the second metatarsal **212**.

(22) FIG. **3A** is a side view of foot **200** showing a typical position of first metatarsal **210** in the sagittal plane. The sagittal plane is a plane parallel to the sagittal structure which divides the body into right and left halves. On foot **200**, the sagittal plane is a plane that extends vertically and is perpendicular to an axis extending medially to laterally along the length of the foot. FIG. **3A** shows first metatarsal **210** with a typical rotational position in the sagittal plane. FIG. **3B** shows first metatarsal **210** with a sagittal plane rotational deformity characterized by a rotational angle **240** relative to ground, as indicated by line **238**.

(23) Bone positioning techniques and instruments according to the disclosure can be useful to correct an anatomical misalignment of a bones or bones. In some applications, the technique involves realigning a portion of a metatarsal relative to an adjacent metatarsal portion. The metatarsal undergoing realignment may be anatomically misaligned in the frontal plane, transverse plane, and/or sagittal plane, as illustrated and discussed with respect to FIGS. **1-3** above.

Accordingly, realignment may involve cutting the metatarsal into two independently movable portions and thereafter realigning one metatarsal portion relative to the other metatarsal portion in one or more planes, two or more planes, or all three planes. After suitably realigning the metatarsal portions, the metatarsal portions can be fixated to hold and maintain the realigned positioned.

(24) While a metatarsal can have a variety of anatomically aligned and misaligned positions, in some examples, the term “anatomically aligned position” means that an angle of a long axis of first metatarsal **210** or portion thereof relative to the long axis of second metatarsal **212** is about 10 degrees or less in the transverse plane and/or sagittal plane. In certain embodiments, anatomical misalignment can be corrected in both the transverse plane and the frontal plane. In the transverse plane, a normal IMA **234** between first metatarsal **210** or portion thereof and second metatarsal **212** is less than about 9 degrees. An IMA **234** of between about 9 degrees and about 13 degrees is considered a mild misalignment of the first metatarsal and the second metatarsal. An IMA **234** of greater than about 16 degrees is considered a severe misalignment of the first metatarsal and the

second metatarsal. In some embodiments, methods according to the disclosure are utilized to anatomically align first metatarsal **210** or a portion thereof by reducing the IMA from over 10 degrees to about 10 degrees or less (e.g., to an IMA of about 1-5 degrees), including to negative angles of about -5 degrees or until interference with the second metatarsal, by positioning the first metatarsal or portion thereof at a different angle with respect to the second metatarsal.

(25) With respect to the frontal plane, a normal first metatarsal will be positioned such that its crista prominence is generally perpendicular to the ground and/or its sesamoid bones are generally parallel to the ground and positioned under the metatarsal. This position can be defined as a metatarsal rotation of 0 degrees. In a misaligned first metatarsal, the metatarsal may be axially rotated between about 4 degrees to about 30 degrees or more. In some embodiments, methods according to the disclosure are utilized to anatomically align the metatarsal or portion thereof by reducing the metatarsal rotation from about 4 degrees or more to less than 4 degrees (e.g., to about 0 to 2 degrees) by rotating the metatarsal or portion thereof with respect to the medial cuneiform.

(26) FIG. **4** is a flow diagram illustrating an example osteotomy technique for correcting an anatomical alignment. The technique will be described with respect to first metatarsal **210** although can be performed on other bones, as discussed above. For purposes of discussion, the technique of FIG. **4** will be discussed with respect to two different images of first metatarsal **210** illustrated in FIGS. **5A** and **5B** to show how different cuts can be made along the first metatarsal during a bone correction procedure. FIG. **5A** is sectional image of first metatarsal **210** from the frontal plane of the bone showing example cut lines that can be made while performing the technique of FIG. **4**. FIG. **5B** is a perspective image of first metatarsal **210** from FIG. **5A**.

(27) With reference to FIGS. **4** and **5**, the example technique involves making a first longitudinal cut **350** into first metatarsal **210** (**300**) and also making a second longitudinal cut **352** into the first metatarsal that intersects the first longitudinal cut (**302**). The first longitudinal cut **350** may be made along the length of first metatarsal **210** between the proximal end **354** and distal end **356**. In addition, the first longitudinal cut **350** may be made through the medial side **358** and the lateral side **360** of first metatarsal **210**, thereby transecting the metatarsal along the lateral plane. In some examples, first longitudinal cut **350** is made from the medial side **358** of first metatarsal **210** toward the lateral side **360** of the metatarsal. The clinician may insert a cutting instrument from the medial side **358** of the bone and guide the cutting instrument towards the lateral side **360** of the bone to cut in this direction. Alternatively, the clinician may insert the cutting instrument from the lateral side **360** of the bone and guide the cutting instrument towards the medial side **358** of the bone. This alternative cutting direction may be useful when performing the procedure on, for example, the fifth metatarsal.

(28) The clinician can make the first longitudinal cut **350** from the proximal end **354** of first metatarsal **210** toward the distal end **356** of the metatarsal. Alternatively, first longitudinal cut **350** can be made from the distal end **356** of the first metatarsal **210** toward the proximal end **354** of the bone. In some examples, first longitudinal cut **350** does not extend the entire length of first metatarsal **210** but instead extends only along a portion of the length of the metatarsal. For example, the proximal end of first longitudinal cut **350** may be offset from the proximal-most end **354** of first metatarsal **210** a separation distance **361**. The distal end of first longitudinal cut **350** may also be offset from the distal-most end **356** of first metatarsal **210** a separation distance **362**.

(29) In some examples, separation distance **361** and separation distance **362** are substantially equal, e.g., such that first longitudinal cut **350** is substantially centered along the length of first metatarsal **210**. In other examples, separation distance **361** and separation distance **362** are different from each other, e.g., such that first longitudinal cut **350** extends more proximally along the length of first metatarsal **210** then distally along the length. In some examples, first longitudinal cut **350** extends along at least 50% of the overall length of first metatarsal **210** between the proximal-most end **354** and the distal-most end **356** such as, e.g., at least 60% of the overall length, or from 50% to 80% of the overall length.

(30) The technique of FIG. 4 also includes making second longitudinal cut 352 into first metatarsal 210 (302). The second longitudinal cut 352 may be made along the length of first metatarsal 210 between the proximal end 354 and the distal end 356. The second longitudinal cut 352 may extend the same length along first metatarsal 210 as first longitudinal cut 350 or may have a different length. Similarly, the second longitudinal cut 352 may be made from the medial side 358 of first metatarsal 210 or the lateral side 360 of the metatarsal, as discussed above with respect to the first longitudinal cut 350. In either case, the second longitudinal cut 352 may be made at a converging angle with respect to the first longitudinal cut 350 to form a wedge-shaped cut.

(31) In the example of FIGS. 5A and 5B, second longitudinal cut 352 is shown as being radially offset from first longitudinal cut 350 (e.g., on the medial side 358 of first metatarsal 210) about an angle 364. In particular, in the illustrated example, first longitudinal cut 350 extends in a common transverse plane across first metatarsal 210 (in the medial to lateral direction) without angling in a plantar or dorsal direction across the bone. By contrast, second longitudinal cut 352 extends across first metatarsal 210 (in the medial to lateral direction) at an angle such that the medial end of second longitudinal cut 352 is positioned closer to the dorsal side of the metatarsal than the lateral side of the cut and, correspondingly, the lateral side of the cut is positioned closer to the plantar side of the metatarsal than the medial side of the cut. In the illustrated arrangement, second longitudinal cut 352 intersects first longitudinal cut 350 at the lateral side 360 of the first metatarsal 210, e.g., such that the apex of the resulting wedge is the lateral side of the bone. In other applications, second longitudinal cut 352 may not intersect first longitudinal cut 350 at the lateral side 360 of the first metatarsal 210 but may instead intersect the first longitudinal cut closer to the medial side 358 of the bone and/or may extend through the lateral side 360 of the first metatarsal without intersecting the first longitudinal cut.

(32) Accordingly, in various applications, the first longitudinal cut 350 into first metatarsal 210 and/or the second longitudinal cut 352 into first metatarsal 210 may or may not extend all the way through the metatarsal (e.g., in the transverse plane). In FIGS. 5A and 5B, first longitudinal cut 350 and second longitudinal cut 352 are illustrated as both extending through first metatarsal 210 such that the cuts intersect at or outside of the lateral side 360 of first metatarsal 210. However, one of the cuts (e.g., which may be designated the first longitudinal cut) may extend through first metatarsal 210 while the other of the cuts extends into the first metatarsal and intersects the through cut within the cross-section of the bone. Thus, it should be appreciated that while first longitudinal cut 350 and second longitudinal cut 352 are illustrated as both extending through first metatarsal 210, one of the cuts may be an intersecting cut within the bone without extending through the entire cross-section of the bone.

(33) In some examples, first longitudinal cut 350 is radially offset from second longitudinal cut 352 by an angle 364 ranging from 10° to 50°, such as from 15° to 35°. In addition, while FIG. 5A illustrates first longitudinal cut 350 extending across first metatarsal 210 in a single transverse plane and second longitudinal cut 352 extending across the first metatarsal at an angle in the dorsal to plantar direction from the medial to lateral side, it should be appreciated that one or both of the longitudinal cuts may be angled (e.g., in the plantar to dorsal direction) across one or more planes (e.g., moving proximally to distally along the first metatarsal and/or moving medially to laterally across the first metatarsal). Therefore, the example cut configuration of FIG. 5A is for purposes of illustration, and the disclosure is not limited in this respect.

(34) To release the bone wedge formed by first longitudinal cut 350 and second longitudinal cut 352, the example technique of FIG. 4 also includes making a first transverse cut (304) and a second transverse cut (306). For example, FIG. 5B illustrates a bone wedge 365 formed by making a first transverse cut 366 and a second transverse cut 368. One or both of the first transverse cut 366 and the second transverse cut 368 may be made before or after making the first longitudinal cut 350 and/or making the second longitudinal cut 352. In general, the specific cutting order in which the first longitudinal cut 350, the second longitudinal cut 352, the first transverse cut 366, and the

second transverse cut **368** are made can be rearranged at the discretion of the clinician performing the procedure.

(35) The first transverse cut **366** may cut first metatarsal **210** in a frontal plane adjacent the proximal end **354** of the metatarsal. The second transverse cut **368** may cut first metatarsal **210** in a different frontal plane adjacent the distal end **356** of the metatarsal. In some examples, such as the example illustrated in FIG. 5B, the first transverse cut **366** extends downwardly, e.g., from a dorsal to plantar direction, from second longitudinal cut **352** and intersects both the first longitudinal cut **350** and the second longitudinal cut **352**. Similarly, the second transverse cut **368** can extend upwardly, e.g., from a plantar to dorsal direction, from first longitudinal cut **350** and intersect both the first longitudinal cut **350** and the second longitudinal cut **352**. This configuration of cuts can form a distal first metatarsal portion **370** whose length is defined by the dorsal portion of first metatarsal **210** and a proximal first metatarsal portion **372** whose length is defined by the proximal portion of the first metatarsal **210**. Alternatively, the configuration of the first transverse cut **366** and the second transverse cut **368** can be reversed such that the first transverse cut extends upwardly from the first longitudinal cut **350** and the second transverse cut **368** extends downwardly from the second longitudinal cut **352**. In this alternative configuration, the distal first metatarsal portion **370** can have a length defined by the plantar portion of the first metatarsal **210** while the proximal first metatarsal portion **372** can have a length defined by the dorsal portion of the metatarsal.

(36) Independent of the direction in which the first transverse cut **366** and the second transverse cut **368** are made, one or both of the cuts may be within a single frontal plane or may be angled along the length of the first metatarsal across multiple frontal planes. For example, first transverse cut **366** may extend distally away from the end of second longitudinal cut **352**. Second transverse cut **368** may also extend distally away from the end of the second longitudinal cut **352**. This configuration of angled transverse cuts can provide a lock and key configuration, such as an interlocking “Z” shape, to help restrict relative movement between the distal first metatarsal portion **370** and the proximal first metatarsal portion **372** following realignment of the two bone portions. In alternative configurations, first transverse cut **366** may be angled proximally from the end of the second longitudinal cut **352**, and second transverse cut **368** may be angled proximally from the end of the second longitudinal cut **352**. In yet further configurations, the first transverse cut **366** or the second transverse cut **368** can be angled proximally while the other transverse cut is orthogonal or angled distally.

(37) The example technique of FIG. 4 also involves moving the distal portion **370** of the first metatarsal **210** relative to the proximal portion **372** of the metatarsal in one or more planes to realign the distal portion of the bone relative the proximal portion (**308**). This can adjust the anatomical alignment of the distal portion **370** of the metatarsal relative to the proximal portion of the metatarsal, e.g., to help correct an anatomical misalignment. In some examples, the distal portion **370** of the first metatarsal **210** is rotated relative to the proximal portion **372** of the metatarsal. The distal portion **370** of the first metatarsal **210** may be rotated in the frontal plane and/or pivoted in the transverse plane and/or pivoted in the sagittal plane to help correct an anatomical misalignment of the distal portion **370**. In some applications, the bone is pivoted by translating and/or sliding the bone. In some examples, the distal portion **370** of the first metatarsal **210** is rotated about an axis extending through the frontal plane so the medial side is moved dorsally and/or the distal portion **370** of the first metatarsal **210** is moved laterally in the transverse plane and/or plantarly in the sagittal plane. For example, the distal portion **370** of the first metatarsal **210** may be moved from an anatomically misaligned position relative to the proximal portion **372** of the first metatarsal and/or the second metatarsal **212** and/or the medial cuneiform **222** to an anatomically aligned position.

(38) To facilitate movement of the distal portion **370** of the first metatarsal relative to the proximal portion **372** of the metatarsal, the wedge **365** formed by cutting the metatarsal can be removed from

the space between the two bone portions. For example, the wedge-shaped bone portion **365** may be removed from between the two bone portions and the bone portions distracted or separated from each other to facilitate relative realignment. Alternatively, the wedge-shaped bone portion **365** may be removed from between the two bone portions and the bone portions realigned relative to each other without distracting the bone portions. In either case, the bone portions can shift and move relative to each other during realignment.

(39) Wedge **365** removed from first metatarsal **210** may or may not be reused as an autograft to fill space created between distal portion **370** and proximal portion **372** of first metatarsal **210** during realignment. FIG. 5C is a sectional view of first metatarsal **210** from the frontal plane showing an example use of wedge-shaped bone portion **365** to close a gap created during realignment of distal portion **370** relative to proximal portion **372**. FIG. 5D is a dorsal perspective view of the first metatarsal **210** with wedge-shaped bone portion **365** from FIG. 5C.

(40) In the example shown in FIGS. 5C and 5D, wedge-shaped bone portion **365** has been removed from the medial side **350** of first metatarsal **210**. This creates a wedge-shaped gap on the medial side between distal portion **370** and proximal portion **372**. During subsequent realignment of distal portion **370** relative to proximal portion **372**, the gap or void space created by removing wedge-shaped bone portion **365** can be closed. For example, as distal portion **370** is rotated in the frontal plane and/or translated in the transverse plane, the gap created by removing wedge-shaped bone portion **365** may be partially or fully closed. As the gap created by removing wedge-shaped bone portion **365** is closed, a second gap may be created on an opposite side of the first metatarsal. This gap or void space created by realignment of the two bone portions may be left open for self-healing or, as illustrated, a bone implant may be inserted into the gap to promote accelerated and efficacious recovery.

(41) In some examples, wedge-shaped bone portion **365** defines a wedge extending from a base **359** to an apex **369**. When resected from the medial side of first metatarsal **210**, base **359** may be on the medial half of the medial half of the metatarsal while apex **369** is on the lateral half of the metatarsal. After being resected from first metatarsal, wedge-shaped bone portion **365** may be rotated 180 degrees (e.g., in the frontal plane and/or transverse plane) so apex **369** is oriented medially and base **359** is oriented laterally. The rotated wedge can then be inserted into the gap created through realignment between the distal portion **370** and proximal portion **372**. The clinician may trim or otherwise resize the wedge-shaped bone portion **365** to be configured (e.g., sized and/or shaped) to fit within the opening created during realignment between the distal portion **370** and proximal portion **372**. After trimming, the bone portion **365** inserted into the opening created by realignment between distal portion **370** and proximal portion **372** may or may not have a wedge shape (e.g., tapering from a wider base to a narrower apex). It should be appreciated that the bone portion **365** may be inserted while the gap is opening and/or after distracting distal portion **370** and proximal portion **372**. Accordingly, realignment between distal portion **370** and proximal portion **372** need not be performed or complete before inserting bone portion **365** on an opposite side of the metatarsal from which the bone portion was extracted.

(42) The bone portion inserted in the opening created by realignment between distal portion **370** and proximal portion **372** may extend partially or fully across the cross-section of first metatarsal **210** in the transverse plane. In the example of FIG. 5C, bone portion **365** is illustrated as extending across the entire width of first metatarsal **210**, e.g., such that the bone portion **365** is positioned between distal portion **370** and proximal portion **372** on both the medial side **350** of the metatarsal and the lateral side **360** of the metatarsal. However, bone portion **365** may be inserted only part way across the cross-section of first metatarsal **210**, e.g., such that distal portion **370** and proximal portion **372** contact each other on the medial side **350** of the metatarsal.

(43) When the wedge-shaped bone portion **365** resected from first metatarsal **210** is reused as a bone implant, the bone portion becomes an autograft within the surgical technique. Instead of reusing the bone portion harvested from the first metatarsal as an autograph, the clinician may use a

different bone construct to fill the gap created through realignment, such as an allograft harvested from another person, a xenograft harvested from a different species, or synthetic bone.

(44) After suitably moving the two transected bone portions relative to each other and, if desired, inserting a bone construct between the two bone portions, the bone portions (and optional inserted bone member) can be fixated to each other to secure and hold the new realigned position achieved through movement (**310**). In some examples, the distal and proximal bone portions (and optional inserted bone member) are provisionally fixated relative to each other before permanently fixating the bone portions relative to each other. Provisional fixation can temporarily hold the proximal bone portion **372** and distal bone portion **370** in fixed alignment relative to each other while one or more permanent fixation devices are applied to the bones and across the joints formed therebetween. A fixation wire and/or a compression pin, such as a threaded olive pin, may be used as provisional fixation instruments.

(45) Independent of whether the distal bone portion **370** and proximal bone portion **372** are provisionally fixated together, the clinician may apply a permanent fixation device to the bone portions and across the joint between the bone portions (and optional inserted bone member). The permanent fixation device can hold the bone portions in fixed alignment relative to each other, e.g., to promote healing between the bone portions in their aligned positions. In different examples, one or more bone plates, pins, screws, staples, or other fixation mechanisms can be used to fixate the bones relative to each other.

(46) In some examples, the clinician may resect bone around the perimeter of the cut end of the distal portion **370** of the first metatarsal and/or the proximal portion **372** of the first metatarsal. Rotating the distal portion **370** of the first metatarsal relative to the proximal portion **372** of the first metatarsal may cause the end of one of the bone portions to rotate out of plane, creating a projecting lip or edge. The clinician may use a cutting instrument to resect the lip or edge of the out-of-plane bone, helping to create a flush surface at the joint between the bone portions. The clinician may resect the protruding lip or edge of the cut end of the bone portion after moving the proximal and distal bone portions relative to each other. In different examples, the clinician can perform such resection before fixating the bone portions relative to each other or after fixating the bone portions relative to each other (e.g., either provisionally or permanently).

(47) While the foregoing description of an example technique involves making a first longitudinal cut **350**, a second longitudinal cut **352**, a first transverse cut **366**, and a second transverse cut **368**, alternative implementations of the technique may be performed by making a lesser number of cuts. As one example, first metatarsal **210** may be separated into a first portion and second portion by making two longitudinal cuts extending along the length of the metatarsal. The longitudinal cuts can extend along the length of the first metatarsal through the proximal end and distal end of the metatarsal. This can separate the first metatarsal **210** into two portions and release a removable bone wedge.

(48) FIGS. 5E and 5F are a medial side view and a frontal plane view, respectively, of first metatarsal **210** illustrating an example technique in which the metatarsal is separated into two portions by making two cuts along the length of the metatarsal. As shown, first longitudinal cut **350** extends through the proximal end **354** and the distal end **356** of the first metatarsal **210** and also through the medial side **358** and lateral side **360** of the metatarsal. Second longitudinal cut **352** is radially offset from first longitudinal cut **350** (e.g., on the medial side **358** of first metatarsal **210**) about angle **364**. Second longitudinal cut **352** also extends through the proximal end **354** and the distal end **356** of the first metatarsal **210** as well as through the medial side **358** and lateral side **360** of the metatarsal.

(49) The angle **364** of offset between first longitudinal cut **350** and second longitudinal cut **352** may be within the ranges discussed above with respect to FIGS. 5A and 5B. For example, first longitudinal cut **350** may be radially offset from second longitudinal cut **352** by an angle **364** ranging from 10° to 50°, such as from 15° to 35°. In addition, in the illustrated example, first

longitudinal cut **350** and second longitudinal cut **352** each extend transversely across first metatarsal **210** (in the medial to lateral direction) without angling in a plantar or dorsal direction across the bone. However, one or both of the longitudinal cuts may angle, e.g., in a planar to dorsal direction or vice versa across the length of the bone, as discussed above.

(50) In practice, a clinician can use a cutting instrument to cut first metatarsal **210** along first longitudinal cut **350** and second longitudinal cut **352**. This can separate the first metatarsal into a first portion **363**, a second portion **367**, and wedge-shaped bone portion **365**. The wedge-shaped bone portion **365** formed by cutting the metatarsal can be removed from the space between the two bone portions. For example, the wedge-shaped bone portion **365** may be removed from between the two bone portions and the bone portions distracted or separated from each other to facilitate relative realignment. Alternatively, the wedge-shaped bone portion **365** may be removed from between the two bone portions and the bone portions realigned relative to each other without distracting the bone portions. In either execution, the bone portions can be realigned relative to each other, as discussed above.

(51) In some examples, the wedge-shaped bone portion **365** removed from between the first bone portion **363** and the second bone portion **367** is used as a bone implant in a gap during realignment of the first bone portion **363** relative to the second bone portion **367**. For example, as discussed with respect to FIGS. 5C and 5D, a gap created by removing wedge-shaped bone portion **365** may be closed while realigning the first bone portion **363** relative to the second bone portion **367**. As this gap is closed, a second gap may be created on an opposite side of the first metatarsal. The wedge-shaped bone portion **365** resected from first metatarsal **210** can be reused as a bone implant in this second gap. For example, FIGS. 5H and 5G are a medial side view and a frontal plane view, respectively, showing an example use of wedge-shaped bone portion **365** in a metatarsal realignment technique. As discussed above with respect to FIGS. 5C and 5D, wedge-shaped bone portion **365** may be rotated 180 degrees (e.g., in the frontal plane and/or transverse plane) and inserted into the gap created through realignment between the first bone portion **363** and the second bone portion **367**. Instead of reusing the bone portion harvested from the first metatarsal as an autograph in the technique of FIGS. 5E and 5H, the clinician may use a different bone construct to fill the gap created through realignment, such as an allograft harvested from another person, a xenograft harvested from a different species, or synthetic bone.

(52) After suitably moving the two bone portions relative to each other and, if desired, inserting a bone construct between the two bone portions, the bone portions (and optional inserted bone member) can be fixated to each other to secure and hold the new realigned position achieved through movement. In some examples, the bone portions (and optional inserted bone member) are provisionally fixated relative followed by permanent fixation.

(53) A variety of different cutting instruments and cutting guides can be used to implement the technique discussed above with respect to FIGS. 4 and 5. FIG. 6A is a perspective view of one example cutting guide **400** that can be used to cut a first metatarsal **210** to realign a distal portion **370** of the metatarsal relative to a proximal portion **372**. As shown in the illustrated example of FIG. 6A, cutting guide may be defined by a body **402** that is configured (e.g., sized and/or shaped) to be positioned against a bone to be cut. For example, body **402** may define a front side **404A** and a backside **404B** that is opposite the front side. Body **402** may further extend from a first longitudinal side edge **406A** to a second longitudinal side edge **406B**. In use, body **402** may be positioned against a first metatarsal to be cut such that backside **404B** is in contact with the metatarsal while the front side **404A** faces outwardly away from the metatarsal. When so positioned, first longitudinal side edge **406A** can be positioned closer to the proximal end of the first metatarsal from the distal end, while second longitudinal side edge **406B** can be positioned closer to the distal end of the first metatarsal and the proximal end.

(54) Body **402** of cutting guide **400** may have a variety of different shapes. In some examples, body **402** is shaped to conform to the curvature of a bone to be cut. For example, body **402** may have a

“V” shape or a radius of curvature that conforms to the radius of curvature of a bone to be cut. When so configured, body **402** may wrap at least partially about the curved external surface of the bone to be cut.

(55) To cut a first metatarsal along the length of the bone, cutting guide **400** can have at least one longitudinal cutting slot **408**. The longitudinal cutting slot **408** may extend at least partially, and in some examples fully, along the length of the cutting guide body **402** between the first longitudinal side edge **406A** and the second longitudinal side edge **406B**. The longitudinal cutting slot **408** may extend across body **402** at a constant vertical location on the body (e.g., as illustrated in FIG. **6A**) or may slope upwardly or downwardly across the length of the body.

(56) To create transverse cuts on the first metatarsal, cutting guide **400** may further include at least one transverse cutting slot which, in the example of FIG. **6A**, is illustrated as two transverse cutting slots: first transverse cutting slot **410** and second transverse cutting slot **412**. First transverse cutting slot **410** may extend downwardly from the longitudinal cutting slot **408**, and the second transverse cutting slot **412** may extend upwardly from the longitudinal cutting slot. For example, longitudinal cutting slot **408** may run from a first terminal end **414** to a second terminal end **416**. First transverse cutting slot **410** can extend downwardly from the first terminal end **414** of the longitudinal cutting slot, e.g., such that the first transverse cutting slot and the longitudinal cutting slot share a common terminal end **414**. Second transverse cutting slot **412** can extend upwardly from the second terminal end **416** of the longitudinal cutting slot, e.g., such that the second transverse cutting slot in longitudinal should cutting slot share a common terminal end **416**.

(57) The first transverse cutting slot **410** can intersect the longitudinal cutting slot **408** at a first intersection angle **418**. The second transverse cutting slot **412** can intersect the longitudinal cutting slot **408** at a second intersection angle **420**. In the illustrated example, the first and second intersection angles **418**, **420** are illustrated as being approximately orthogonal or 90°. In different examples, the first and second intersection angles **418**, **420** may range from 20° to 135°, such as less than 90°, from 20° to 80°, from 30° to 75°, or yet other angles. In some examples, the first and second intersection angles **418**, **420** may be offset from orthogonal from 1° to 45°. The first and second intersection angles **418**, **420** may be the same or may be different from each other.

(58) In some examples, first transverse cutting slot **410** is sloped proximately moving away from longitudinal cutting slot **408**. For example, when cutting guide **400** is applied to a first metatarsal, terminal end **414** of the first transverse cutting slot may be positioned more distally along the length of the bone than the opposite end of the cutting slot. Where second transverse cutting slot **412** is sloped at a corresponding angle, terminal end **416** of the second transverse cutting slot may be positioned more proximately along the length of the bone than the opposite end of the cutting slot.

(59) Bone cutting guide **400** can have any suitable dimensions. In some examples, longitudinal cutting slot **408**, first transverse cutting slot **410**, and second transverse cutting slot **412** each have a width ranging from 0.1 mm to 3 mm. The dimensions of each of the different cutting slots may be the same or may be different from each other. For example, the dimensions of each of the cutting slots may be the same and may be sized based on the particular cutting instrument (e.g., saw blade, rotary bur, osteotome) intended to be used for the surgical procedure. Longitudinal cutting slot **408**, first transverse cutting slot **410**, and/or second transverse cutting slot **412** may extend perpendicularly through the thickness of body **402** or may be angled as the slot extends through the thickness of the body, e.g., to provide an angled cutting slot.

(60) To attach bone cutting guide **400** to a bone to be cut during a surgical procedure, body **402** may include one or more fixation apertures configured to receive a fixation member, such as a wire, pin, screw, or other mechanical fixation element intended to temporarily secure and hold the cutting guide to the bone during the surgical procedure. In the example of FIG. **6A**, body **402** includes a first slot **422** and a second slot **424** that function as fixation apertures. First slot **422** is positioned between the first longitudinal edge **406A** of body **402** and first transverse cutting slot

410. Second slot **424** is positioned between the second longitudinal side edge **406B** of body **402** and second transverse cutting slot **412**. First slot **422** and second slot **424** are illustrated as being parallel to first transverse cutting slot **410** and second transverse cutting slot **412**, respectively, but may not be parallel in other configurations. In general, slots **422** and **424** may be sufficiently long to allow cutting guide **400** to be positioned at different radial or angular locations about the first metatarsal **210** being cut, e.g., to allow different cuts at different radial or angular positions along the bone to be made using longitudinal cutting slot **408**.

(61) FIG. 6B is a perspective view of cutting guide **400** attached to a first metatarsal **210** using slots **422** and **424**. As shown in this example, fixation pins **426** are inserted through each of slots **422** and **424** to position bone cutting guide **400** at a first radial position on the first metatarsal. Bone cutting guide **400** can be rotated radially about first metatarsal **210** to a second radial position while fixation pins **426** remain in slots **422** and **424**, e.g., to reposition the location of longitudinal cutting slot **408**. This can be useful to position bone cutting guide **400** at one location to make first longitudinal cut **350** (FIG. 5A) and subsequently rotate the cutting guide to a second location to make second longitudinal cut **352**. Bone cutting guide **400** may include one or more additional fixation apertures **428A**, **428B** to secure the bone cutting guide at a particular rotational orientation about first metatarsal **210**.

(62) FIG. 7A is a perspective illustration of another example configuration of bone cutting guide **400** where like features discussed above with respect to FIGS. 6A and 6B refer to like elements. As shown in this example, bone cutting guide **400** includes the previously described longitudinal cutting slot **408**, first transverse cutting slot **410**, and second transverse cutting slot **412**. In addition, in the example configuration of FIG. 7A, bone cutting guide **400** includes a second longitudinal cutting slot **440**. Second longitudinal cutting slot **440** may extend parallel to the first longitudinal cutting slot **408** and may be offset relative to the first longitudinal cutting slot. For example, the first longitudinal cutting slot **408** and the second longitudinal cutting slot **440** may be offset from their centerlines a distance **442** ranging from 2 mm to 35 mm, such as from 5 mm to 20 mm.

(63) Second longitudinal cutting slot **440** can extend parallel to first longitudinal cutting slot **408** and intersect first transverse cutting slot **410**. For example, second longitudinal cutting slot **440** may intersect first transverse cutting slot **410** at a location between the terminal ends of the transverse cutting slot. In use, a clinician may use first longitudinal cutting slot **408** and second longitudinal cutting slot **440** to make the two longitudinal cuts along the length of first metatarsal **210** discussed above with respect to FIGS. 4 and 5. In such applications, first longitudinal cutting slot **408** and second longitudinal cutting slot **440** may be angled relative to each other to form converging cut lines that intersect on first metatarsal **210**. For example, first longitudinal cutting slot **408** and second longitudinal cutting slot **440** may be angled relative to each other to form cutting lines that converge at an angle ranging from 5° to 45°.

(64) To help form second transverse cut **368** (FIG. 5B), cutting guide **400** in FIG. 7A may include a third transverse cutting slot **444** that is co-linear with a second transverse cutting slot **412**. For example, second longitudinal cutting slot **440** may extend from a first end **445** to a second end **448**. The first end **445** of the second longitudinal cutting slot **440** can intersect first transverse cutting slot **410**. Third transverse cutting slot **444** can extend upwardly from the second end **448** of the second longitudinal cutting slot **440**. In practice, a clinician may insert a cutting instrument through both second transverse cutting slot **412** and third transverse cutting slot **444** to form a joined, single transverse cut line **368** (FIG. 5B) to help separate a distal metatarsal portion **370** from a proximal metatarsal portion **372**.

(65) To help control the depth of cut being made by a clinician using bone cutting guide **400**, the cutting guide may include a depth limiter. The depth limiter may be a surface against which the clinician can position a cutting instrument and along which the clinician can translate the cutting instrument through a cutting slot in order to set the depth of cut made through the slot. The depth

limiter may be permanently joined/integrally formed with the body **402** or may be a separate component attachable to or used in conjunction with body **402**.

(66) In the example of FIG. **7A**, bone cutting guide **400** includes a depth limiter **446**, which is defined by a surface projecting outwardly from the first side **404A** of body **402**. In particular, in the illustrated configuration, depth limiter **446** includes one surface or rail projecting outwardly from body **402** on one side of second longitudinal cutting slot **440** and a second surface or rail projecting outwardly from the body on an opposite side of the longitudinal cutting slot. In some examples, depth limiter **446** projects outwardly from the first side **404A** of body **402** a distance ranging from 0.25 mm to 10 mm. A clinician may place their cutting instrument against depth limiter **446**, e.g., with a cutting blade or member inserted through the slot, and guide the cutting instrument along the depth limiter to form a cut of controlled depth.

(67) As discussed above with respect to the example configuration of bone cutting guide **400** in FIGS. **6A** and **6B**, bone cutting guide **400** in the example of FIG. **7** may include one or more fixation apertures to temporarily fixate the bone cutting guide against a bone to be cut during a procedure. In the illustrated configuration, bone cutting guide **400** includes three fixation apertures **428**, although a different number of fixation apertures may be used. FIG. **7B** is a perspective illustration of bone cutting guide **400** from FIG. **7A** showing the cutting guide attached to an example first metatarsal **210**.

(68) As mentioned, a variety of different cutting guide configurations can be used during a bone realignment procedure according to the present disclosure. FIG. **8** is a perspective illustration of another example bone cutting guide **500** that can be used according to the disclosure. As shown in this example, bone cutting guide **500** has a body **502** that extends from a first end **504** to a second end **506**. A longitudinal cutting slot **508** is formed through the body **502** and extends along the length of the body between the first end **504** and the second end **506**. A clinician may manipulate cutting guide **500** to position longitudinal cutting slot **508** at different positions along a first metatarsal **210** to be cut.

(69) To temporarily secure cutting guide **500** at one or more different positions against the bone to be cut, the cutting guide may include securing projections. For example, cutting guide **500** may include a first securing projection **510A** and a second securing projection **510B** extending substantially orthogonally (or at a different angle) relative to the length of the body. Each securing projection can have at least one, and in some examples multiple, fixation apertures configured to receive a mechanical fixation element for securing the body to a bone to be cut. For example, each securing projection may include multiple fixation apertures that are linearly aligned (e.g., in an orthogonal direction away from the length of the body). The different securing projections may allow bone cutting guide **500** to be positioned at different degrees of rotation about the bone to be cut, e.g., by selecting different rotational positions corresponding to different sets of fixation apertures.

(70) In different examples, bone cutting guide **500** may include more than two securing projections, such as three, four or more securing projections. In the illustrated configuration, bone cutting guide **500** includes four securing projections **510A-510D**. First and second securing projections **510A** and **510B** may be located adjacent the first end **504** of body **502**, while the third and fourth securing projections **510C** and **510D** may be located adjacent the second end **506** of the body.

(71) To manipulate bone cutting guide **500** during use, the bone cutting guide may include a handle **512**. The handle may extend from the body **502** and angle ranging from 10° to 90°. The handle can be permanently attached to body **502** or can be removably coupled to the body, e.g., threadingly coupled to the body. In some applications, a clinician may hold bone cutting guide **500** against a bone to be cut while performing a cutting operation without securing the cutting guide to the bone with a mechanical fixation instrument using a securing projection. In other applications, the clinician may manipulate bone cutting guide **500** against the bone using handle **512** but may provisionally fixate the bone cutting guide against the bone using a mechanical fixation element

inserted through one or more securing projections before performing a cutting procedure. FIGS.

9A-9D are conceptual illustrations showing how bone cutting guide **500** can be manipulated relative to a bone to be cut to execute a bone realignment technique as discussed herein.

(72) In the foregoing description, certain features, elements, and techniques have been described with respect to relative references such as “upwardly” and “downwardly.” It should be appreciated that such relative references are used for purposes of illustration based on the orientation in the figures. More generally, the features, elements, and techniques may change orientation in three-dimensional space. Accordingly, references to “upwardly” and “downwardly” should be considered relative and not confined to a specific orientation with respect to gravity.

(73) Various examples have been described. These and other examples are within the scope of the following claims.

Claims

1. A method of correcting a bunion, the method comprising: positioning a cutting guide against a medial side of a first metatarsal, the cutting guide having a body defining a backside positioned against the first metatarsal, a frontside positioned facing away from the first metatarsal, a first side edge, and a second side edge, the second side edge being positioned closer to the distal end of the first metatarsal than the first side edge, wherein the cutting guide comprises an angled cutting slot having a first terminal end and a second terminal end with the first terminal end positioned more distally along a length of the first metatarsal than the second terminal end; guiding a cutting instrument through the angled cutting slot to cut the first metatarsal into a distal metatarsal portion and a proximal metatarsal portion; moving the distal metatarsal portion relative to the proximal metatarsal portion in a transverse plane and in a frontal plane to correct a bunion misalignment of the first metatarsal; applying provisional fixation to temporarily hold the distal metatarsal bone portion in a fixated alignment relative to the proximal metatarsal bone portion; and applying permanent fixation to hold the distal metatarsal bone portion in a fixated alignment relative to the proximal metatarsal bone portion.
2. The method of claim 1, wherein moving the distal metatarsal portion in the frontal plane comprises rotating the distal metatarsal portion until a tibial sesamoid bone and a fibular sesamoid bone are on opposite sides of a sagittal plane when viewed from the frontal plane.
3. The method of claim 1, wherein moving the distal metatarsal portion in the transverse plane comprises reducing an intermetatarsal angle between the distal metatarsal portion and a second metatarsal to 10 degrees or less.
4. The method of claim 1, wherein moving the distal metatarsal portion in the transverse plane comprises reducing an intermetatarsal angle between the distal metatarsal portion and a second metatarsal to within 1 to 5 degrees.
5. The method of claim 1, wherein guiding the cutting instrument through the angled cutting slot to cut the first metatarsal comprises cutting the first metatarsal at a proximal portion of the first metatarsal.
6. The method of claim 1, wherein the cutting guide comprises fixation apertures, and further comprising, prior to cutting the first metatarsal, attaching the cutting guide to the first metatarsal proximally and distally of a location where the first metatarsal is to be cut.
7. The method of claim 6, wherein the fixation apertures comprises slots.
8. The method of claim 1, wherein applying permanent fixation comprises applying one or more of a bone plate, pin, screw, and staple.
9. The method of claim 1, wherein applying permanent fixation comprises applying a bone plate and screws.
10. The method of claim 1, wherein applying permanent fixation comprises applying one or more staples.

11. The method of claim 1, wherein applying provisional fixation comprises applying a fixation wire and/or a compression pin between the proximal metatarsal portion and the distal metatarsal portion.
 12. The method of claim 1, wherein guiding the cutting instrument through the angled cutting slot to cut the first metatarsal further comprises cutting a wedge-shaped section of bone from the first metatarsal.
 13. The method of claim 1, wherein the angled cutting slot has a width within a range from 0.1 mm to 3 mm.
 14. The method of claim 1, wherein the angled cutting slot is a first transverse cutting slot, and the cutting guide further comprises a second transverse cutting slot and a longitudinal cutting slot extending from the first transverse cutting slot to the second transverse cutting slot.
 15. The method of claim 1, wherein guiding the cutting instrument through the angled cutting slot to cut the first metatarsal comprises making a first transverse cut across the first metatarsal, and further comprises making a second transverse cut with the cutting instrument across the first metatarsal.
 16. The method of claim 1, wherein the cutting guide comprises a handle extending outwardly away from the body, and positioning the cutting guide against the medial side of the first metatarsal comprises manipulating the cutting guide with the handle.
 17. The method of claim 1, wherein the cutting instrument comprises a saw blade.
 18. The method of claim 1, wherein the cutting instrument comprises a burr.
 19. The method of claim 1, wherein guiding the cutting instrument through the angled cutting slot to cut the first metatarsal comprises guiding the cutting instrument from the medial side of the first metatarsal through a lateral side of the first metatarsal.
 20. The method of claim 1, wherein the angled cutting slot comprises a first transverse cutting slot, and the cutting guide further comprises a second transverse cutting slot offset from the first transverse cutting slot along a length of the first metatarsal.
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